# María José Briceño Nikulin

## Backend Software Engineer · Developer Consultant at Thoughtworks

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## PROFILE

Backend Software Engineer working with Java, Spring Boot and hexagonal architecture, deploying to Kubernetes and GCP with GitLab CI/CD and monitoring with Grafana and Prometheus. Electrical Engineer with a background in industrial automation and process control; experience in intelligent transportation, electronic voting and digital government. Diplomas in Artificial Intelligence and Software Engineering (Universidad de Chile).

## TECHNOLOGIES

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|---|---|
| **Languages** | Java (primary), Python, Scala, JavaScript |
| **Frameworks** | Spring Boot, Django, Play |
| **Architecture** | Hexagonal architecture (ports and adapters), microservices |
| **Cloud & CI/CD** | Kubernetes, Google Cloud Platform (GCP), GitLab CI/CD |
| **Observability** | Grafana, Prometheus |
| **Databases** | Oracle, SQL, MongoDB |

## WORK EXPERIENCE

**Developer Consultant** — Thoughtworks — Jan 2025 – present

- Backend development with Java, Spring Boot and hexagonal architecture
- Deployment and operation on Kubernetes and GCP, with GitLab CI/CD pipelines
- Observability with Grafana and Prometheus

**Engineering Consultant** — Independent — Jul 2024 – May 2025

- Software development consulting with Java

**Software Engineer** — SUBTEL, Ministry of Transport and Telecommunications of Chile — Feb 2023 – Jul 2024

- Technical Lead for implementing the Digital Government Law at SUBTEL
- Development, support and maintenance of the platforms supporting SUBTEL's operations
- Design and implementation of new features for a microservices-based technology upgrade

**Software Engineer** — EVoting Global — Mar 2022 – Dec 2022

- Development and maintenance of features for the EHolders (shareholder meetings) and EVoters (elections software) platforms

**Software Engineer** — Sonda S.A. — Sep 2016 – Mar 2022

*Fleet Management Project: on-board console software*

- Design, driver development and functional testing for GPS, passenger counters and fare validators
- Design and modeling of algorithms to locate a bus along its route, estimate arrival times at upcoming stops, and locate it inside a tunnel
- Electrical schematics for on-board circuitry; automated file loading into the on-board equipment's local database
- Field test design and execution; support to Testing and Operations for testing and production releases
- Historical data analysis for decision-making (Pandas, TensorFlow)
- Contributed to a patent associated with the product; production rollout of two prototypes (fare-evasion measurement and Transantiago equipment certification)

*KMS Project*

- Design and development of an application to manage the security of transit system cards

## EDUCATION

**Diploma in Software Engineering**, Dept. of Computer Science, Universidad de Chile — Highest distinction — 2024

**UCR INNOVA'R I-Corps-Based Training**, University of California, Riverside — 2020

**Diploma in Artificial Intelligence**, Dept. of Computer Science, Universidad de Chile — Highest distinction — 2019 – 2020

**Master of Engineering Sciences, Electrical Engineering**, Universidad de Chile (voluntary withdrawal) — 2015 – 2017

**Electrical Engineer** (Ingeniera Civil en Electricidad), Universidad de Santiago de Chile — 2013

## PUBLICATIONS

Widzyk-Capehart, E., Barberán, A., **Briceño, M. J.**, Navarro, C., Nguyen, P. M. V., Opazo, C., & Steffen, S. (2019). *Collocated ground deformation and pore pressure measurements in open pit mines: laboratory testing and analysis of wireless sensing platform*. MPES 2018, Springer, pp. 381–391.

## LANGUAGES

Spanish (native) · English (professional working) · Portuguese (basic)